

Pangmiaomiao Zhang

Ph.D. Candidate in Chemistry (expected May 2027)

 Austin, TX
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Qualification

- Analytical chemist specializing in bioanalytical assay development and nucleic acid-protein interactions.
- Experienced in designing biochemical and single-molecule workflows, quantitative characterization, instrument troubleshooting, mentoring, and interdisciplinary collaboration.

Education

Ph.D. The University of Texas at Austin, Chemistry candi- • Expected graduation: May 2027 date • GPA: 3.95/4.00	Austin, TX Aug 2022 – May 2027
M.Sc. Tsinghua University, Chemistry • GPA: 3.87/4.00	China Sept 2019 – June 2022
B.Sc. Southern University of Science and Technology, Chemistry • GPA: 3.85/4.00	China Sept 2015 – June 2019

Experience

Lin Bio-AFM Lab, The University of Texas at Austin, Graduate Researcher • Designed DNA substrates and optimized biochemical and single-molecule assays to study G2L4 reverse transcriptase-mediated DNA repair; first-author manuscript accepted in Nature Communications. • Investigated how N- and O-linked glycosylation changes SARS-CoV-2 spike-antibody binding, complex formation, and aggregation using deglycosylation, SDS-PAGE, mass photometry, HS-AFM, and TEM. • Expressed and purified full-length SARS-CoV-2 membrane protein and optimized detergent solubilization, liposome reconstitution, and structural imaging workflows.	Austin, TX 2023 – present 3 years
The University of Texas at Austin, Teaching Assistant • Led chemistry discussion or laboratory sessions and evaluated undergraduate assignments.	Austin, TX 2022 – present 4 years
Lin Bio-AFM Lab, The University of Texas at Austin, HS-AFM Instrument Manager • Managed shared HS-AFM instrumentation, trained new users, troubleshoot performance, and developed standardized biological sample-preparation and imaging protocols.	Austin, TX 2022 – present 4 years
Texas Venture Lab, Associate Consultant • Conducted market research, competitive analysis, and financial modeling for a biotechnology startup developing gene-editing approaches for mosquito population control. • Presented technical and commercial findings to startup stakeholders.	Austin, TX Jan 2025 – May 2025 5 months
Tsinghua University, Graduate Researcher • Developed fluorescence sensor arrays for amyloid beta aggregates and disease-associated urinary protein profiles, including human urine samples in a clinical collaboration. • Developed polymer-peptide fluorescence probes for p300- and SIRT1-related enzyme activity assays.	China 2019 – 2022 3 years

Southern University of Science and Technology, Undergraduate Researcher

China

- Developed asymmetric annulation and catalyst-free cycloaddition reactions using small-molecule synthesis, chromatography, NMR, HPLC, and X-ray crystallography.

2017 – 2018

1 year

Skills

Bioanalytical and Biochemical Assays: PCR, ELISA, enzyme activity assays, fluorescence assays, SDS-PAGE, Western blotting, DNA-PAGE

Molecular Biology and Sample Preparation: oligonucleotide and primer design, cell homogenization, nucleic acid and protein extraction, protein expression and purification, dialysis, membrane protein reconstitution

Analytical Characterization: mass spectrometry, HPLC-MS, mass photometry, HS-AFM, TEM, NMR, CD, fluorescence spectroscopy, UV-Vis spectroscopy, DLS

Data Analysis and Scientific Software: Python, MATLAB, ImageJ/Fiji, ChimeraX, Igor Pro, Origin, MNOVA, ChemDraw

Awards

Chemistry Department Research Fellowship

2026

The University of Texas at Austin

Outstanding M.Sc. Thesis Award and Outstanding Graduate

2022

Tsinghua University

Outstanding B.Sc. Thesis Award and Outstanding Graduate

2019

Southern University of Science and Technology